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MIL-F-44402(GL)
16 February 1990

MILITARY SPECIFICATION

FINISHING SECTION, MODULAR PRINTING SYSTEM

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the finishing section of the Modular Printing System (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 3610

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SPECIFICATIONS

MILITARY

MIL-D-3464 - Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-S-44195 - Shelter, Tactical, Expandable, Two-Side

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Department of the Army Technical Manuals

TM 5-3610-294-13&P - Operator's Unit and Intermediate Direct Support Maintenance Manual including Repair Parts and Special Tools List for Modular Printing System, Module C Finishing Section

(Copies of the Army Technical Manuals are available from the U.S. AG Publications Center, 1655 Woodson Road, St. Louis, MO 63144.)

DRAWINGS

Us. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-13-4699	Finishing Shelter Arrangement
5-13-4733	24,000 BTU Heat Pump
5-13-4763	36,000 BTU Heat Pump
5-13-4993	Paper Cutter
5-13-5001	Stencil, Center of Gravity
5-13-5010	Modifications, Exterior Walls
5-13-5025	Shelter, Outside MODS (Finshing)
5-1.3-5026	Data Plate, Transportation Data
5-13-5029	Data Plate, Identification, Finishing Section

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-EMSS, Natick, MA 01760-5014.)

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2.2 Non-Government publications. The following documents forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA No. 70 - National Electrical Code Handbook

(Application for copies should be addressed to the National Fire Protection Association, 80 Battery March Pike, Quincy, MA 02169.)

(Non-Government standards and other publications are normally available from the organizations that prepare of distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Alternate components. Components offered as equivalent to components specified hereinafter' and on the referenced drawings as a specific manufacturer's part or model number or equal, shall be functionally equal to the manufacturer's model number identified, and of equal or better quality. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other components, unless such modification or change is specifically approved by the contracting officer. Prior to manufacture of the first article, or if none is required, prior to commencing production, the contractor shall submit for the contracting officer's approval, a list identifying each proposed alternate component, together with proof that each listed component complies with requirements specified above. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with the necessary supporting data, authorizes commencement of manufacture but does not relieve the contractor of responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

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3.2.1 Materials and components. The materials and components shall be as specified herein and on the applicable drawings. Materials and components not definitely specified shall be-of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this specification. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3 Design and construction. The design and construction of the finishing section shall be as specified herein, and in Drawing 5-13-4699 and all subsidiary drawings and parts lists pertaining thereto. All wiring shall be in accordance with the provisions of NFPA No. 70.

3.4 Finishing section operation. All electrically operated equipment, including switches, lights. telephone, papercutter, and heat pumps shall function without failure when tested for performance and readability as specified in 4.5.1 and 4.5.2.

3.5 Shelter. The shelter (see 3.8) shall be either desert sand or camouflage green as specified [see 6.2). All interior and exterior surfaces of the shelter shall be cleaned and rehabilitated when necessary to restore the original color and finish.

3.6 Technical manuals. Technical manual, 5-3610-294-13&P, shall be provided with the Finishing Section and shall be located within the cabinet next to the papercutter.

3.7 Marking. In addition to existing markings, each completed shelter shall be marked as specified in 3.7.1 and 3.7.2, and as shown on applicable drawings.

3.7.1 End item identification plates. Identification plates and Drawings 5-13-5026 and 5-13-5029 shall be attached to the exterior of the shelter as specified on Drawing 5-13-5025. Appropriate information (NSN, weights, and serial no.) shall be added in the spaces provided on each dataplate.

3.7.2 Center of gravity. The exterior walls of the shelter shall have a center of gravity stencil, Drawing 5-13-5001, located as shown on Drawing 5-13-5010, to identify the center of gravity for the shelter when loaded.

3.8 Government furnished property (GFP). The GFP indicated in table I will be furnished to the contractor by the Government (see 6.2), and shall be a part of the finishing section.

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TABLE I. Government furnished property

Item No.	Quantity	Nomenclature
1	1	Shelter, Tactical, Expandable, Two-Side, MIL-S-44195 1/
2	1	Paper Cutter, Drawing 5-13-4993
3	1	24,000 BTU Heat Pump, Drawing 5-13-4733
4	1	36,000 BTU Heat Pump, Drawing 5-13-4763

1/ Color shall be as specified in contract or purchase order

3.9 Workmanship. The finish shall be free of blisters, peel, chipped areas or areas of no film. The finish shall be free from smudges, poor touch up. scale, scratches, stain, overspray, sag, imprint, or embedded foreign matter. Threaded fasteners shall not be broken, stripped, or loose. Holes for rivets shall be accurately punched or drilled and any burrs that would prevent proper seating of the rivet shall be removed. Rivets shall completely fill the holes, with full size heads neatly finished and in full contact with the surface of the member, and shall not be loose.

3.9.1 Welders. unless otherwise specified (see 6.2), before assigning any welder to work, the contractor shall provide the Government inspection service with the names of the welders to be employed in the work, together with certification that each of the welders has passed a qualification test as prescribed by either of the following listed codes for the type of welding operation to be performed, and that such qualification is effective as defined by that particular code as follows:

a. Standard Qualification Procedure of the American Welding Society.
b. Welding Qualification of the American Society of Mechanical Engineers,

3.9.2 Welding. The surface of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be continuous, sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits (when flux is used) shall be removed from the finished welds, and rough edges and projections shall be dressed smooth.

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3.9.3 Electrical wiring. Wiring shall not be cut, abraded, or have excessive insulation stripped, and shall be properly and tightly joined at terminals. Wiring shall have adequate slack to provide strain relief.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material. l–

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified 4.4.4 and 4.4.5, and shall be tested for the characteristics specified in 4.5.1 and 4.5.2. The presence of any nonconformance or failure to pass any test shall be cause for rejection of first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified. or qualified in this specification or applicable purchase document.

4.4.2 In-process dimensional inspection. Dimensional examinations shall be made throughout the fabrication processes to determine conformance to the dimensions-specified on the applicable drawings. Whenever nonconformance is noted, corrections shall be made to the items and lot in-process. Items which cannot be corrected shall be removed from production.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of completely assembled finishing sections. The sample unit shall be one finishing section. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5 for minor defects. Any sample unit found to contain one or more major defects shall be cause for rejection of the lot.

TABLE II. End item visual defects

Examine	Defect	Classification Major	Classification Minor
Finish	Color not as specified.		201
	Blistered, peeled, chipped, or area of no film.		202
	Poor touch-up, orange peel, color separation, smudge, scale, scratched, stained, overspray, sag, imprint, or foreign matter imbedded.		203
Design, construction and workmanship, general (applicable to all components and assemblies)	Any component or part missing or not specified type or size.	101	
	Fractured, split, punctured, torn, dented, creased, deteriorated, or malformed.	102	
	Component misplaced or not in proper alignment.		204
	Component loose.		205
	Sharp burr or edge or silver.		206
	Any operation omitted or not properly performed.	103	
	Any functioning component that is inoperative or will not operate as intended.	104	
	Any functioning component that requires abnormal force to be operated or not interchangeable.		207
	Any adjustable assembly that cannot be adjusted.		208

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TABLE II. End item visual defects - cont'd

Examine	Defect	Classification Major	Classification Minor
Design, construction and workmanship, general (applicable to all components and assemblies) continued	Any assembly not properly adjusted.		209
	Any component not readily accessible for servicing where required.		210
Welding	Missing, incomplete, cracked, fractured, not fused.	105	
	Undercut, not smooth and uniform, or scale or flux deposits not removed.		211
Bolts, nuts, screws, studs and other types of threaded fasteners	Broken, stripped, fractured, or not tight.		212
Rivets (when required)	Broken, not peened or insufficiently peened, or loose.		213
Electrical assembly	Wiring cut, abraded, or not properly installed or joined.	106	
	Adequate slack not provided for wiring to relieve strain.		214

4.4,4 End Item dimensional inspection. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of finishing sections. The sample unit shall be one finishing section. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. Every finish section shall be tested as specified in 4.5.1. Any test failure shall be cause for rejection of the finishing section tested.

4.4.6 Packaging examination. Examination shall be made of each finishing section to determine that preservation, packing, and marking comply with section 5 requirements. Any deviation from specified requirements shall be cause for rejection of the finishing section.

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4.5 Methods of inspection.

4.5.1 Performance test. Each completed finishing section shall be connected to a suitable electric power source, and the heat pumps, paper cutter and electrical system (lights, switches, buttons, and outlets) shall be operated in accordance with applicable technical manuals to insure proper operation. All doors/drawers shall be checked to insure smooth operation. Any electrical or mechanical malfunction shall constitute failure of this test.

4.5.2 Readability. Following successful completion of test specified in 4.5.1 the complete finishing section, with all contents, shall be secured as specified in section 5 and mounted on Government furnished mobilizers and subjected to a 50 mile road test. The vehicle shall be driven at top legal speed, or as road conditions permit, for this type equipment. Not less than 10 miles of the test course shall be unimproved roads such as earth or partially surfaced gravel, and the remainder shall be improved roads such as concrete, asphalt, or macadam. The test shall include not less than 20 quick starts and stops of the vehicle on grades of not less than 5 percent or greater than 10 percent. 50 percent of the starts and stops shall be made on ascending grades and 50 percent on descending grades. At the completion of the test, the finishing section shall be reinspected for the defects listed in table II and retested as specified in 4.5.1. Any nonconformance noted on completion of the road test shall constitute failure of the test.

5. PACKAGING

5.1 Preservation and packing. Preservation, packing, and marking shall be in accordance with applicable drawings and TM 5-3610-294-13&P. Care should be taken to ensure that desiccant conforming to MIL-D-3464, type I, has been loaded into the shelter before closing. A minimum quantity of 500 units of desiccant shall be loaded into each shelter.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The finishing section is intended-for transport to various distribution areas for use in trimming, cutting, and rolling sheets of printed material into separate leaflets and packaging for distribution. The finishing section is one part of the Modular Printing System and is used in a tactical situation where early and rapid distribution of material is required by PSYOPS units.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When first article is required (see 3.1, 4.3, and 6.3).
d. Color of exterior of shelter (see 3.5).
e. Source and delivery of Government furnished property as applicable (see 3.8).
f. When certificate of qualification test for welders is not required (see 3.9.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

PSYOPS
Shelter, expandable
Trim

Custodian:
Army - CL

Preparing activity:
Army - GL

(Project 3610-A362)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-44402(GL)	Finishing Section, Modular Printing System

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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